



Bountiful AI Assistant

AI-Powered Recipe & Meal Planning Assistant

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Problem Statement & Objectives

The Challenge

Cooking at home is increasingly difficult – people struggle with deciding what to cook, managing limited ingredients, avoiding food waste, and finding recipes that match their dietary restrictions and busy schedules.

Our objectives: recommend recipes from available ingredients, generate personalized meal plans, support dietary preferences, provide nutritional insights, reduce food waste, and make cooking easier for everyone.

What is Bountiful AI?

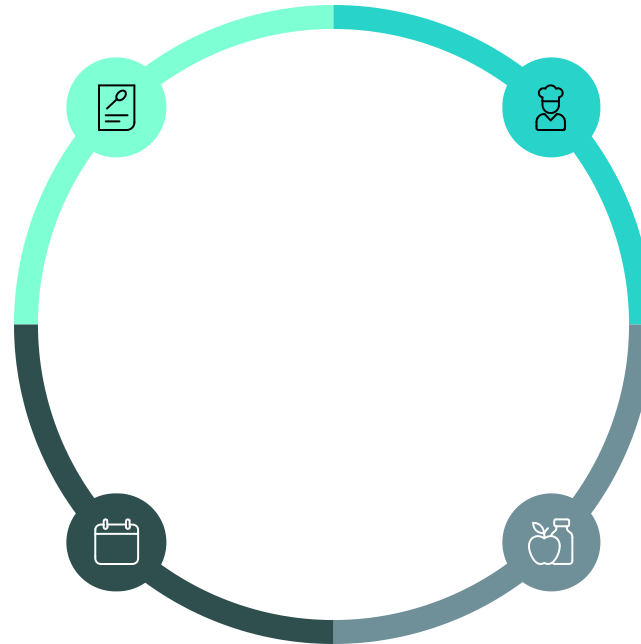
Bountiful AI Assistant is an intelligent recipe and meal-planning assistant powered by AI that helps users prepare delicious and healthy meals with ease.

Recipe Recommendations

AI-driven suggestions tailored to the ingredients you have on hand, eliminating guesswork and saving time.

Personalized Meal Planning

Custom meal plans that align with individual dietary preferences, health goals, and available ingredients.



Step-by-Step Guidance

Detailed cooking instructions and smart ingredient substitutions to help users of all skill levels succeed in the kitchen.

Nutrition Analysis

Comprehensive nutritional insights and healthy eating recommendations for every generated recipe and meal plan.

FEATURES

Key Features

Ingredient Recognition
Identify ingredients from
photos quickly

**Personalized Meal
Planning**
Create plans tailored to
tastes and goals



**AI Recipe
Recommendation**
Suggest recipes using
available items

**Nutrition & Dietary
Support**
Offer nutrition guidance
and diet filters

Every feature is designed
to reduce food waste,
simplify cooking
decisions, and promote
healthier eating habits.



Dietary Preference Support

Supports Vegetarian, Vegan, Gluten-Free, High Protein, and Low Carb preferences so every user gets perfectly matched recipes.



Smart Ingredient Substitutions

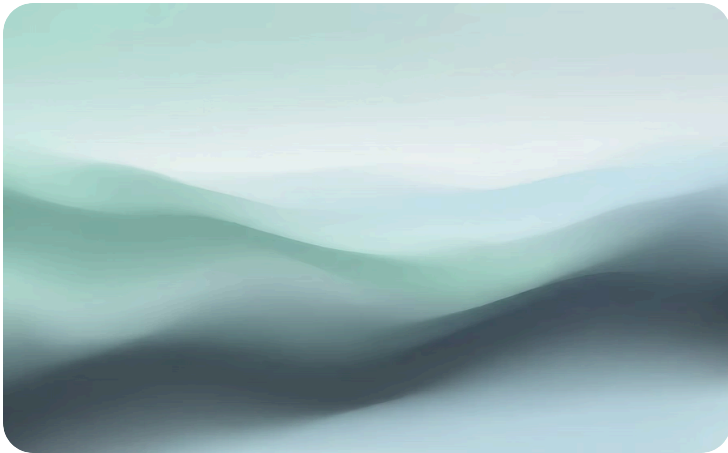
Automatically suggests ingredient alternatives when items are unavailable, reducing waste and keeping meal prep flexible.



Cooking Tips & Best Practices

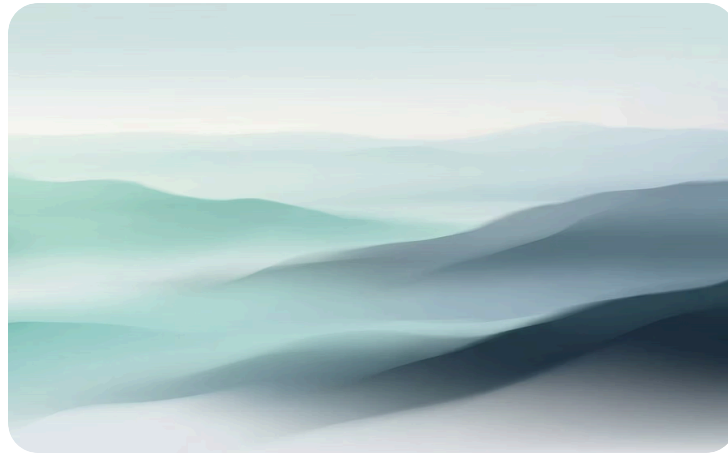
Step-by-step cooking guides and smart tips help beginners and experienced cooks alike elevate every dish.

Technology Stack & Architecture



AI Platform

IBM watsonx.ai with the Granite Foundation Model powers natural language understanding and intelligent recipe generation at the core of Bountiful AI.



Frontend & Backend

A responsive web interface built with HTML, CSS, and JavaScript communicates with a Python/Flask backend to deliver a seamless user experience.



Orchestration & APIs

IBM watsonx Orchestrate manages AI agent workflows, connecting to Recipe & Nutrition APIs and deployed on IBM Cloud for reliability and scale.

Bountiful AI Workflow

End-to-End Intelligence

From the moment a user opens the assistant to receiving a fully personalized recipe and meal plan, every step is powered by IBM watsonx AI.

01

Enter Ingredients or Preferences

The user opens the assistant and inputs available ingredients or specifies dietary preferences and meal goals.

03

Generate Recipe & Nutrition

The system fetches data from Recipe and Nutrition APIs, generating a personalized recipe with full nutritional analysis.

02

AI Understands Intent

IBM watsonx Orchestrate processes the request, understanding user intent through the Granite AI foundation model.

04

Deliver Final Meal Plan

The assistant presents ingredient alternatives, step-by-step cooking guidance, and a complete personalized meal plan.

Welcome back, *Bhuvanesh Purushothaman*

[Collapse](#) ^



Create your agent



Explore agent catalog



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Agents

[Bountiful AI Assistant](#)

Agents

[AskOrchestrate](#)

Needs attention

Alerts

Updated on July 9, 2026 at 3:06 PM [View all](#)

Operations

0

Incidents

0

Insights

2

Alert

Actions

Bountiful AI Assistant has no evaluation runs in the last 30 days



AskOrchestrate has no test cases



Bountiful AI Assistant

AI Model : GPT-OSS 120B – Ope...

- Profile
- Knowledge
- Toolset
- Behavior
- Channels Preview

Behavior

Define how and where your agent should react to requests and respond to users.

Instructions

- Ingredient with quantity

Steps:

1.

2.

3.

Tips:

- Simple cooking or serving tip"

Guidelines

Add rules that guide an agent's actions and responses.

Add Guideline +

Start by creating guidelines

Click **Add Guideline** to create a guideline for the agent.

Build with AITalk to agent

Running in debug mode

Today

You 3:05 PM

Give me recipe of RoseMilk

Bountiful AI Assistant 3:05 PM

Show Reasoning

Rose Milk

Servings: 1

Ingredients:

1 cup (240 ml) whole milk

2 tablespoons rose syrup (or rose water diluted to taste)

1 teaspoon sugar (optional, adjust to sweetness)

Ice cubes (optional)

Steps:

1. Pour the milk into a tall glass.

2. Add the rose syrup (or diluted rose water) and stir well until fully combined.

3. Taste and add sugar if you prefer it sweeter; stir until dissolved.

Save as test

Type something...

ANALYSIS

Benefits, Challenges & Future Scope



Benefits

Saves cooking time, delivers personalized recommendations, supports healthy eating, features a beginner-friendly interface, reduces food waste, and improves overall meal planning.



Challenges

Dependence on external APIs, regional recipe limitations, nutrition data accuracy concerns, and a requirement for internet connectivity present current constraints.



Future Enhancements

Voice assistant, image-based ingredient detection, grocery shopping integration, multilingual support, smart kitchen device integration, and a weekly family meal planner are planned.

CONCLUSION

Conclusion & Thank You

AI-Powered Recipe Assistant

Bountiful AI demonstrates how AI can transform everyday cooking by delivering personalized recipes and intelligent meal planning through IBM watsonx technologies.

Nutrition & Health Insights

The solution promotes healthier eating habits and helps reduce food waste through smart, data-driven recommendations tailored to every user.

Built on IBM watsonx.ai & Orchestrate

Leveraging IBM's enterprise AI ecosystem ensures reliability, scalability, and a user-friendly experience for cooks of every level.

Thank You!